

Chapter 34: Aggregate Demand and Aggregate Supply

Why do recessions happen if economies naturally grow over time?



Learning Objectives

34.1 Explain why the aggregate-demand curve and identify the factors that cause it to shift.

34.2 Differentiate between the short-run and long-run aggregate supply curves and explain what determines their shapes.

34.3 Analyze how sticky-wage and sticky-price theories explain deviations of output from its natural level in the short run.

34.4 Evaluate how economic shocks affect output and prices using the AD-AS model.

34.5 Identify the economy's long-run equilibrium and explain how it is restored after a shock.

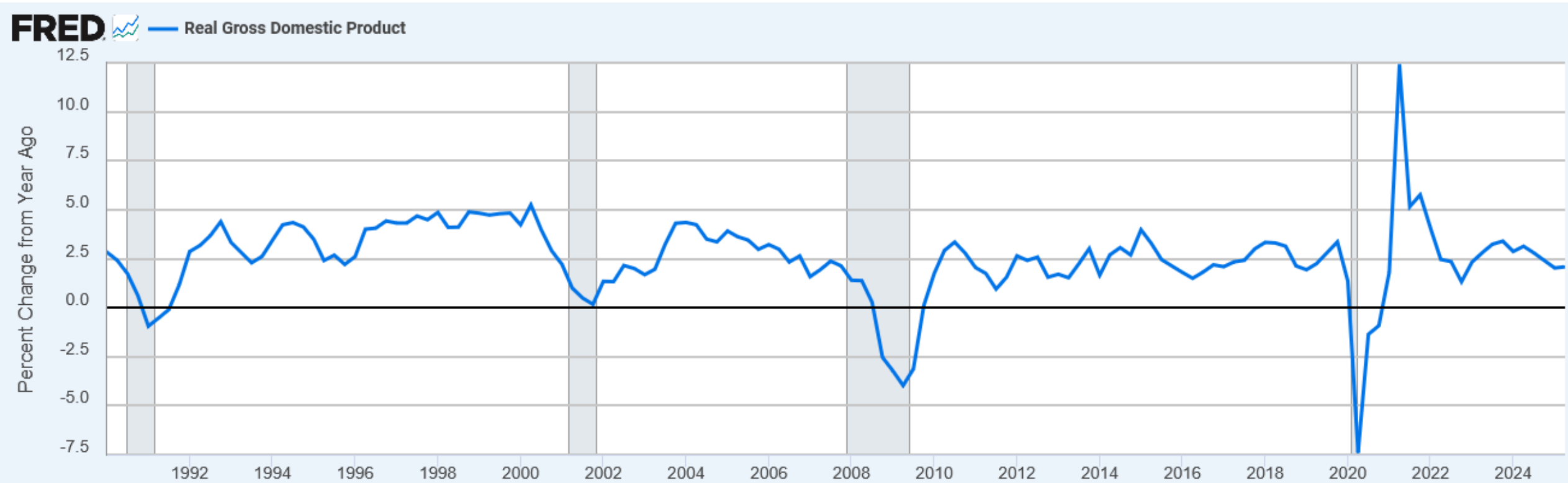
Key Points

- **Economic Fluctuations:** Economies experience unpredictable ups and downs around long-term growth.
- **Classical vs. Short Run:** Classical theory applies in the long run, but in the short run, prices and wages can be sticky—so nominal changes affect real outcomes.
- **Aggregate Demand (AD):** Slopes downward due to the **wealth effect**, **interest-rate effect**, and **exchange-rate effect**.
- **Aggregate Supply (AS):**
 - **Long-run AS (LRAS):** Vertical; output depends on resources, capital, and technology.
 - **Short-run AS (SRAS):** Upward sloping due to sticky wages, sticky prices, and misperceptions.
- **Shifts:**
 - AD shifts with changes in consumption, investment, government spending, or net exports.
 - AS shifts with changes in labor, capital, resources, technology, or expected prices.
- **Economic Shocks:**
 - Demand shocks cause temporary changes in output and prices.
 - Supply shocks can create **stagflation**—rising prices and falling output.

Introduction

- Over the long run, real GDP grows about 3% per year on average.
 - Since 1990: 2.5% with a high of 4.7% in 1999 and a low of -2.8% in 2009.
 - There have only been 8 years with 3%+ growth since 1990
- In the short run, GDP fluctuates around its trend.
 - **Recessions**: periods of falling real incomes and rising unemployment
 - Two consecutive quarters of falling RGDP
 - **Depressions**: severe recessions (very rare)
- Short-run economic fluctuations are often called **business cycles**.

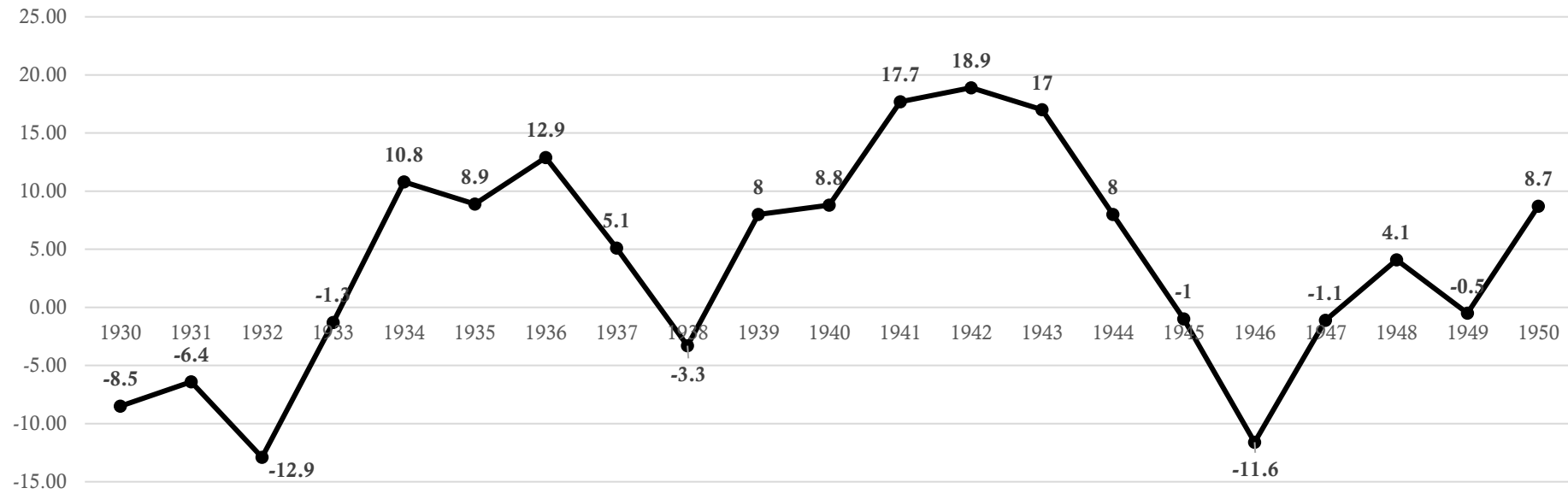
Business Cycles



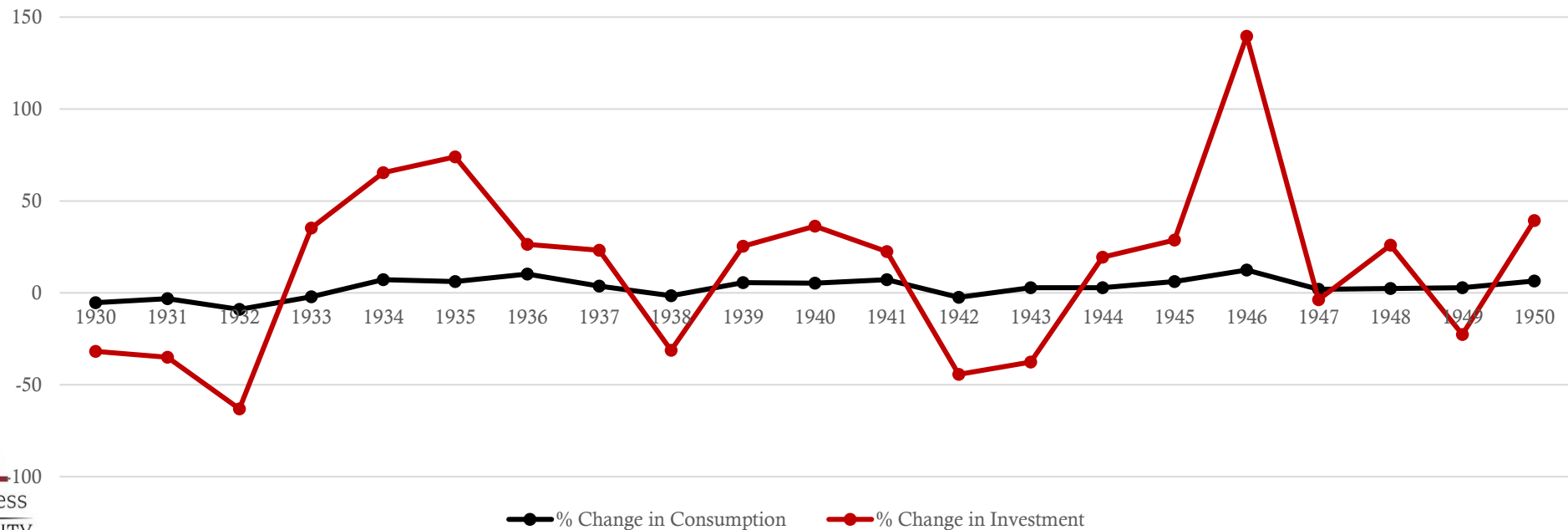
Source: U.S. Bureau of Economic Analysis via FRED®
Shaded areas indicate U.S. recessions.

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% Change in RGDP

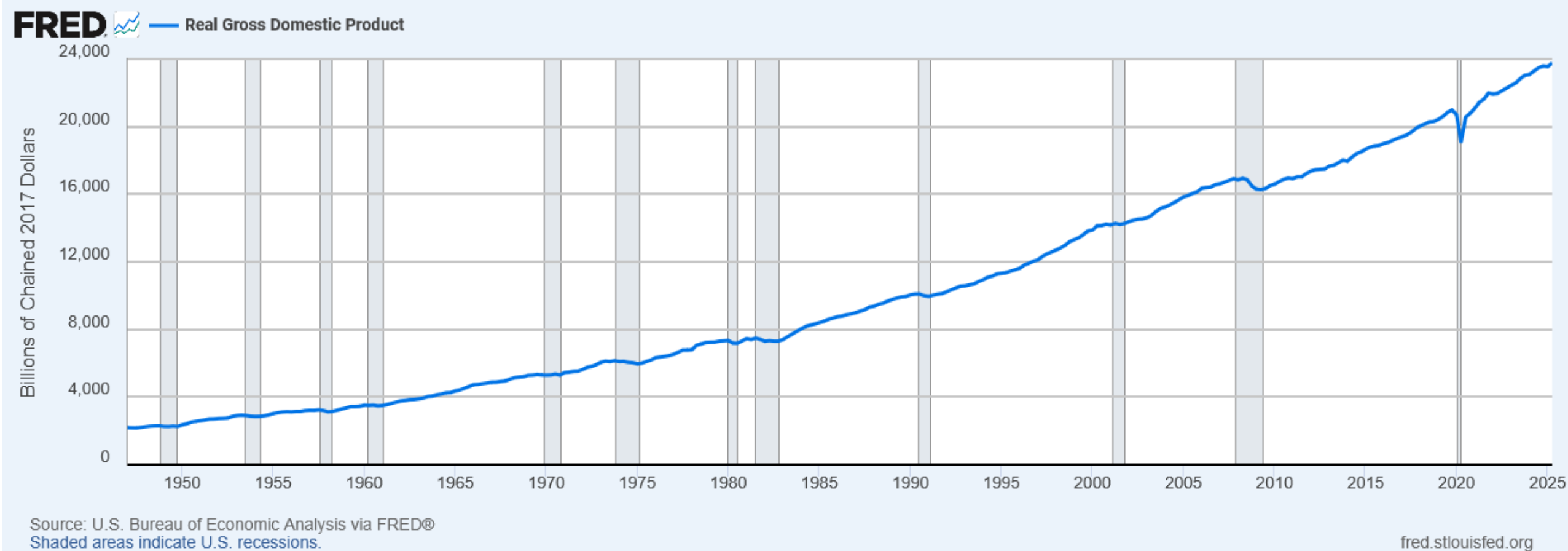


% Change in Consumption and Investment during the Great Depression and WW2 Era



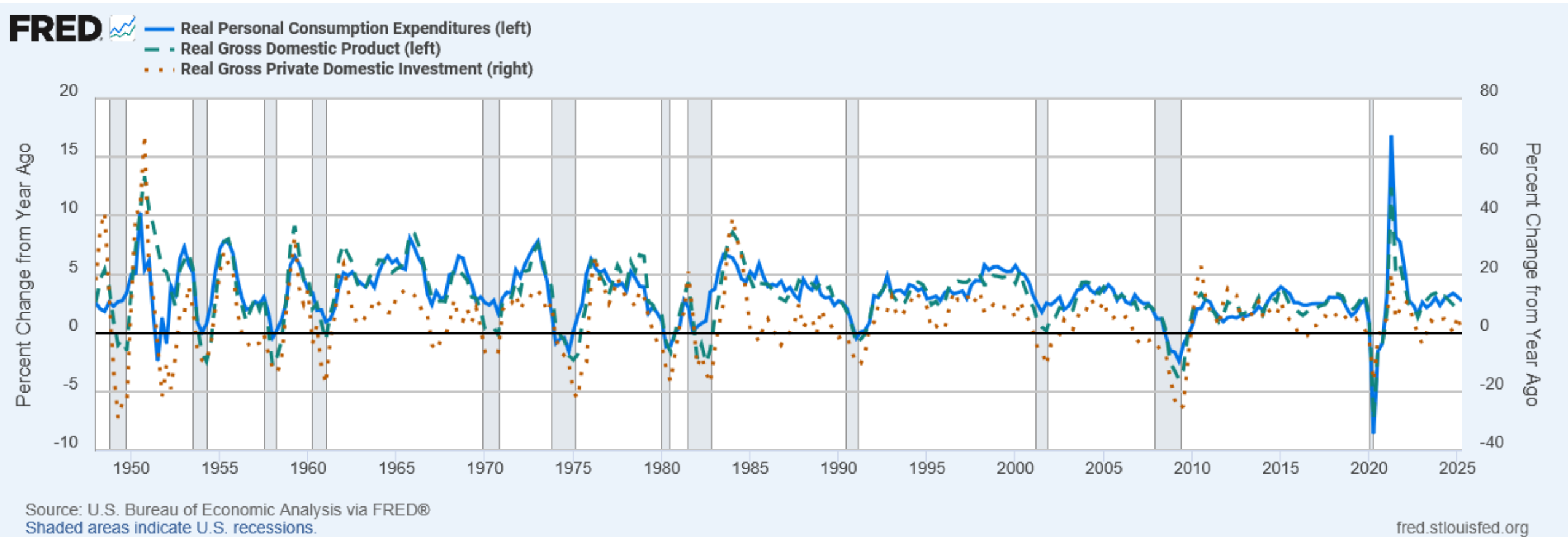
Three Facts about Fluctuations

- **FACT 1:** Economic fluctuations are irregular and unpredictable



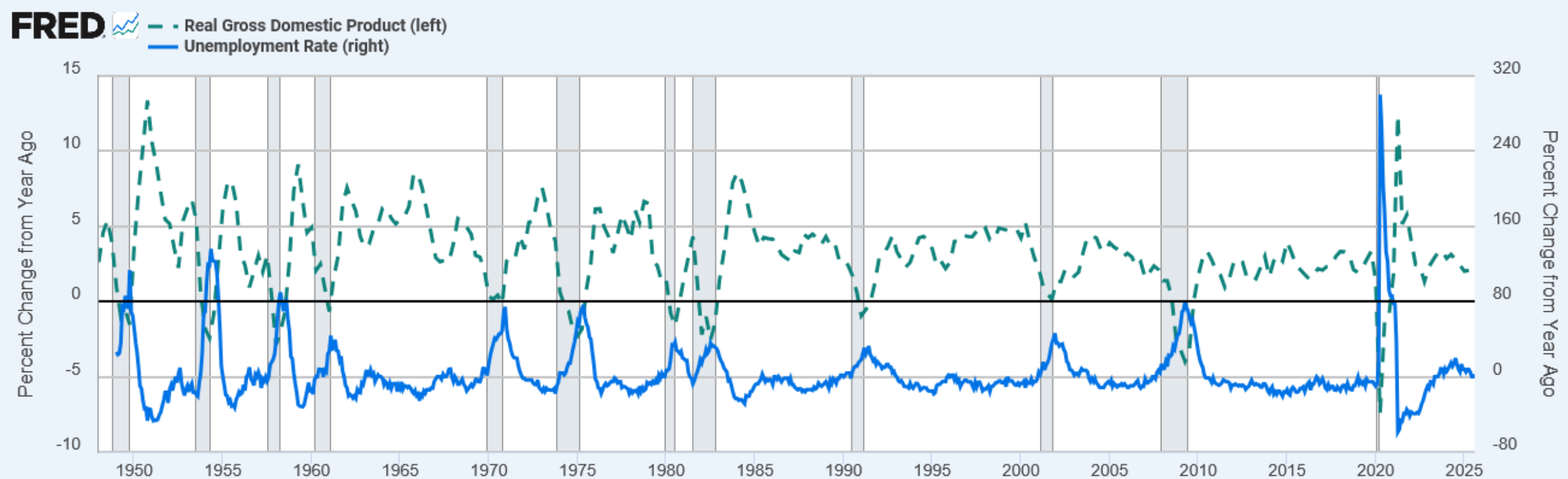
Three Facts about Fluctuations

- **FACT 2:** Most macroeconomic quantities fluctuate together.



Three Facts about Fluctuations

- **FACT 3:** As output falls, unemployment rises.



Sources: U.S. Bureau of Economic Analysis; U.S. Bureau of Labor Statistics via FRED®
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Introduction, *continued*

- Explaining these fluctuations is difficult, and the theory of economic fluctuations is controversial.
- Most economists use the **model of aggregate demand and aggregate supply** to study fluctuations.
- This model differs from the classical economic theories economists use to explain the long run.

Classical Economics—A Recap

- The previous chapters are based on the ideas of classical economics, especially:
- The **Classical Dichotomy**, the separation of variables into two groups:
 - Real – quantities, relative prices
 - Nominal – measured in terms of money
- The **neutrality of money**:
Changes in the money supply affect nominal but not real variables.

Classical Economics

- Most economists believe classical theory describes the world in the long run, but not the short run.
- In the short run, changes in nominal variables (like the money supply or P) can affect real variables (like Y or the u-rate).
- To study the short run, we use a new model.

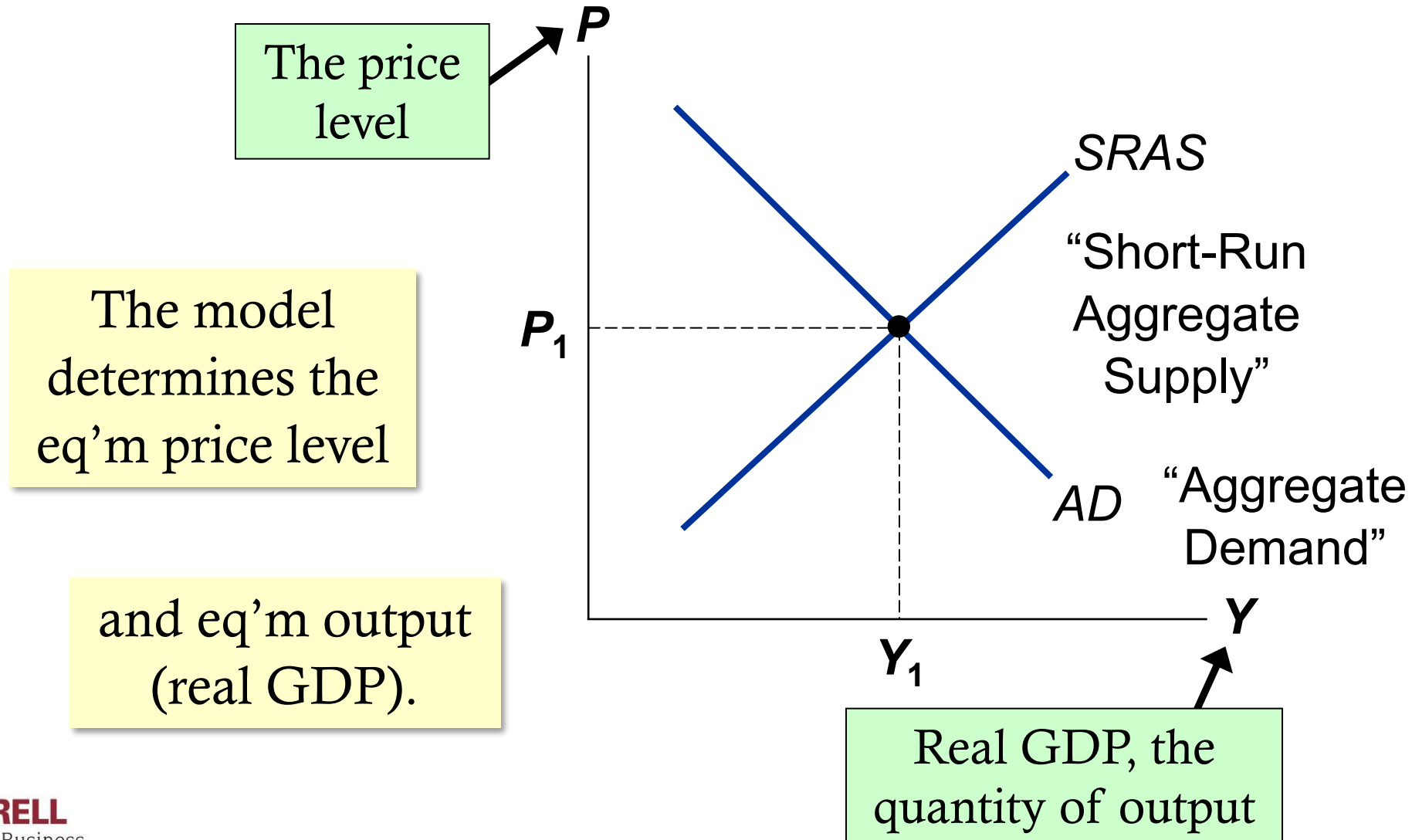
Activity: Spot the Fluctuation

- Below are three recent news headlines. Which describe short-run fluctuations, and which reflect long-run growth trends? Explain your reasoning.
1. “U.S. jobless rate spikes after interest rate hike”
 2. “Tech investment fuels decade of productivity gains”
 3. “Consumer spending dips as confidence falls”

Activity: Spot the Fluctuation

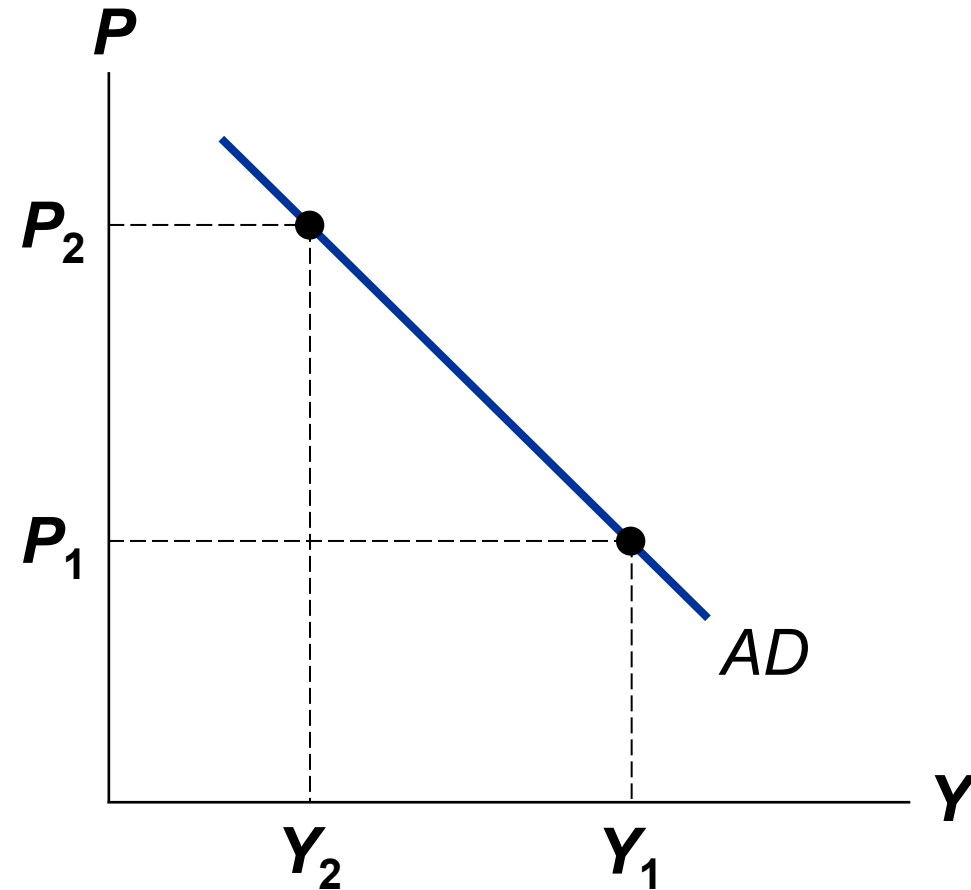
1. **Short-run:** A rise in interest rates reduces borrowing and spending temporarily, leading to lower output and higher unemployment — a movement along the business cycle.
2. **Long-run:** Productivity growth reflects improvements in technology and capital that shift long-run aggregate supply (LRAS) outward — it's a source of long-term economic growth, not a short-run swing.
3. **Short-run:** A decline in consumer confidence leads to lower consumption spending (C), reducing aggregate demand (AD) and causing short-run declines in GDP and employment.

The Model of Aggregate Demand and Aggregate Supply



The Aggregate-Demand (AD) Curve

- The **AD curve** shows the quantity of all goods and services demanded in the economy at any given price level.



Why the AD Curve Slopes Downward

- $Y = C + I + G + NX$
- C = consumption
- I = investment
- NX = net exports (exports – imports)
- Assume G (government expenditures) fixed by government policy.
- To understand the slope of AD , must determine how a change in P affects C , I , and NX .

The Wealth Effect (P and C)

Suppose P rises.

- The dollars people hold buy fewer goods and services, so real wealth is lower.
- People feel poorer.

Result: C falls.

The Interest Rate Effect (P and I)

Suppose P rises.

- Buying goods and services requires more dollars.
- To get these dollars, people sell bonds or other assets.
- This drives up interest rates.
- Result: I falls.
- Recall, I depends negatively on interest rates.
- Businesses spend less on new capital at higher interest rates

The Exchange Rate Effect (P and NX)

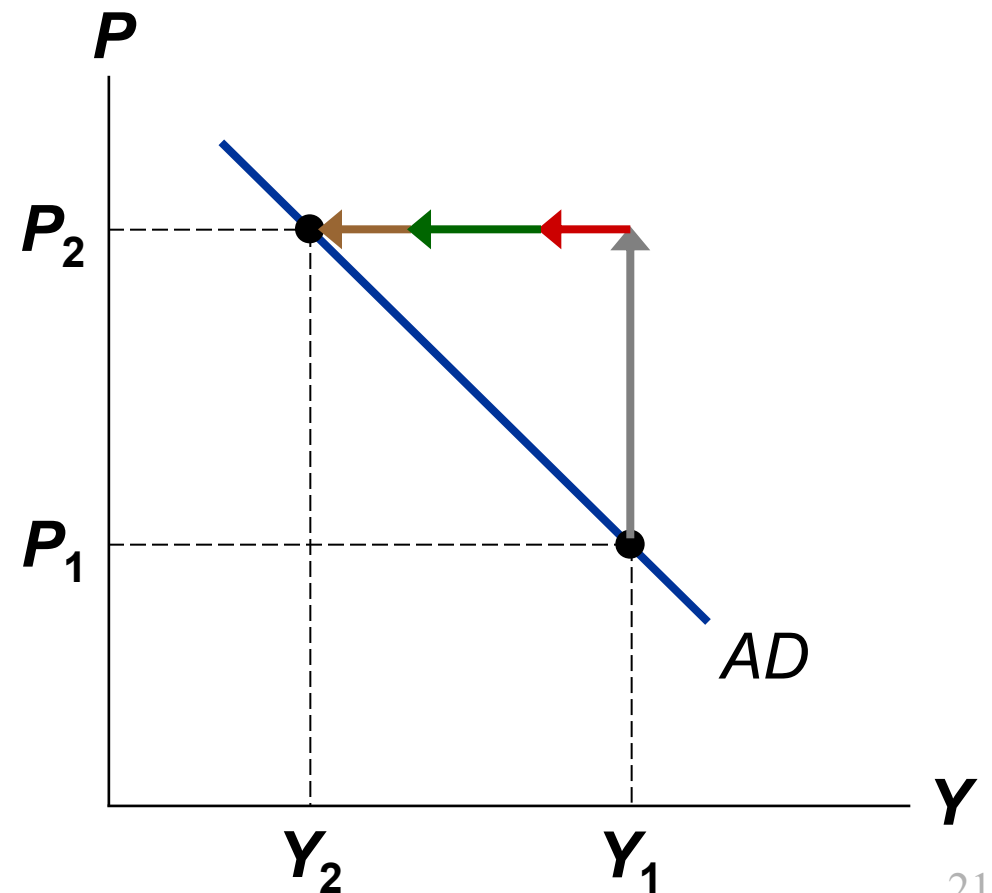
Suppose P rises.

- U.S. interest rates rise (the interest-rate effect).
- Foreign investors desire more U.S. bonds.
- Higher demand for \$ in foreign exchange market.
- U.S. exchange rate appreciates.
- U.S. exports more expensive to people abroad, imports cheaper to U.S. residents.

Result: NX falls.

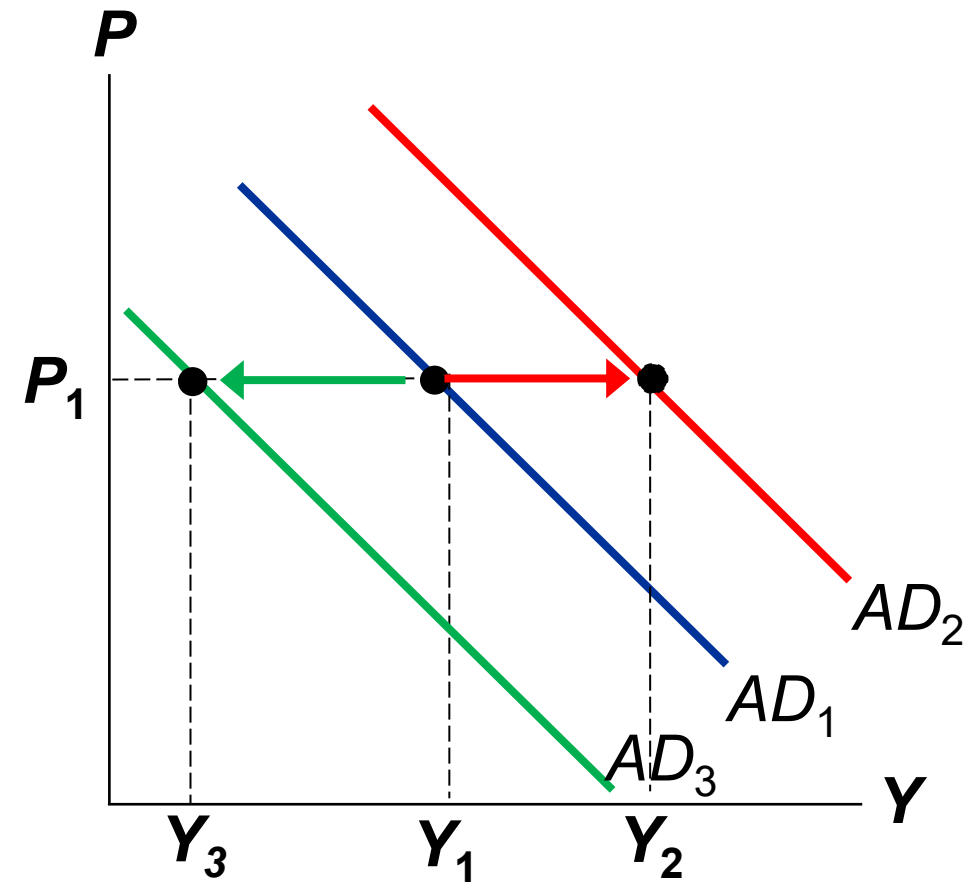
The Slope of the AD Curve: Summary

- An increase in P reduces the quantity of goods and services demanded because:
 - the wealth effect (C falls)
 - the interest-rate effect (I falls)
 - the exchange-rate effect (NX falls)



Why the AD Curve Might Shift

- Any event that changes C , I , G , or NX —except a change in P —will shift the AD curve.
- Example:
A stock market boom makes households feel wealthier, C rises, the AD curve shifts right.
- An increase in taxes causes households to lose wealth, C falls, the AD curve shifts left.



Why the AD Curve Might Shift

- Changes in C
 - Stock market boom/crash (**Increase/Decrease**)
 - Preferences re: consumption/saving tradeoff
 - Tax hikes/cuts (**Decrease/Increase**)
- Changes in I
 - Firms buy new computers, equipment, factories (**Increase**)
 - Expectations, optimism/pessimism (**Increase/Decrease**)
 - Interest rates, monetary policy (**Increase/Decrease**)
 - Investment Tax Credit or other tax incentives (**Increase**)

Why the AD Curve Might Shift

- Changes in G
 - Federal spending, e.g., defense (**Increase/Decrease**)
 - State & local spending, e.g., roads, schools (**Increase/Decrease**)
- Changes in NX
 - Booms/recessions in countries that buy our exports (**Increase/Decrease**)
 - Appreciation/depreciation resulting from international speculation in foreign exchange market (**Decrease/Increase**)

Activity: Shifting the Aggregate Demand Curve

What happens to the *AD* curve in each of the following scenarios?

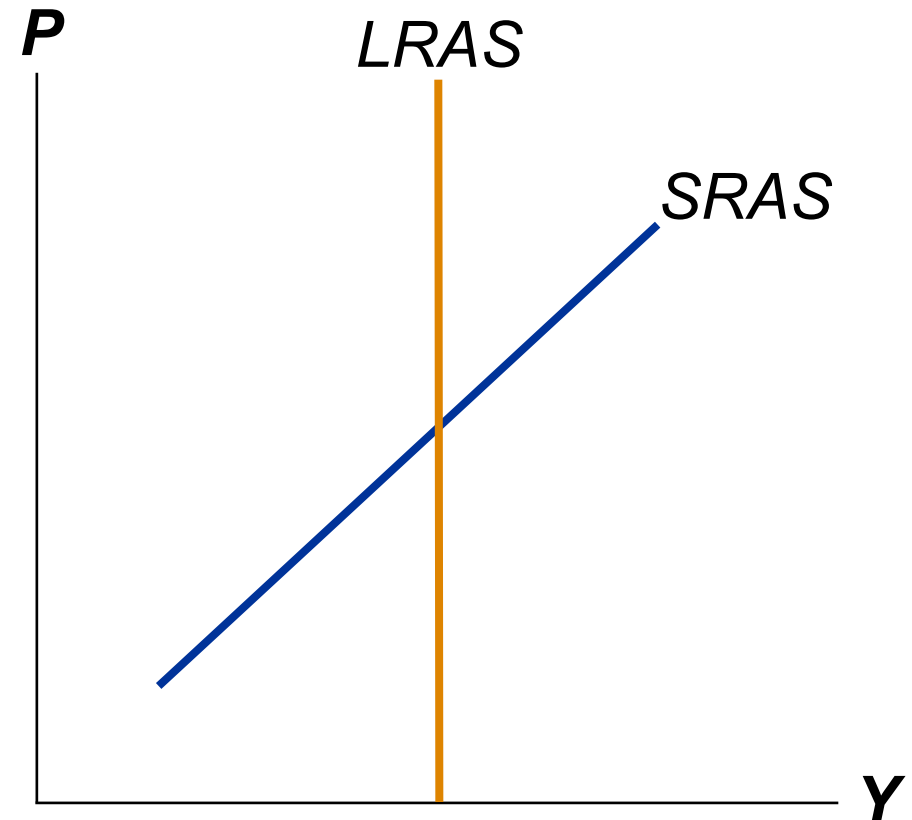
1. A ten-year-old investment tax credit expires.
2. The U.S. exchange rate falls.
3. A fall in prices increases the real value of consumers' wealth.
4. State governments replace their sales taxes with new taxes on interest, dividends, and capital gains.

Activity: Shifting the Aggregate Demand Curve

1. A ten-year-old investment tax credit expires.
 - I falls, AD curve shifts left.
2. The U.S. exchange rate falls.
 - \$ depreciates, NX rises, AD curve shifts right.
3. A fall in prices increases the real value of consumers' wealth.
 - Move down along AD curve (wealth-effect).
4. State governments replace sales taxes with new taxes on interest, dividends, and capital gains.
 - C rises, AD shifts right.

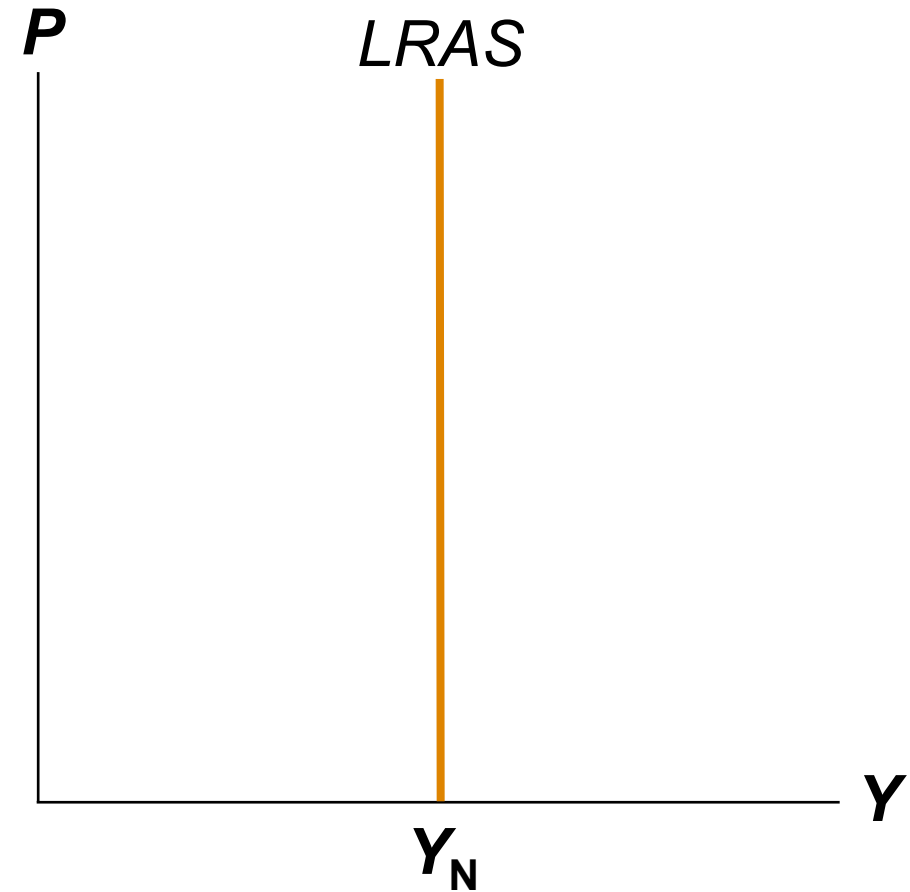
The Aggregate-Supply (*AS*) Curves

- The ***AS* curve** shows the total quantity of goods and services firms produce and sell at any given price level.
- *AS* is:
 - upward-sloping in short run
 - vertical in long run

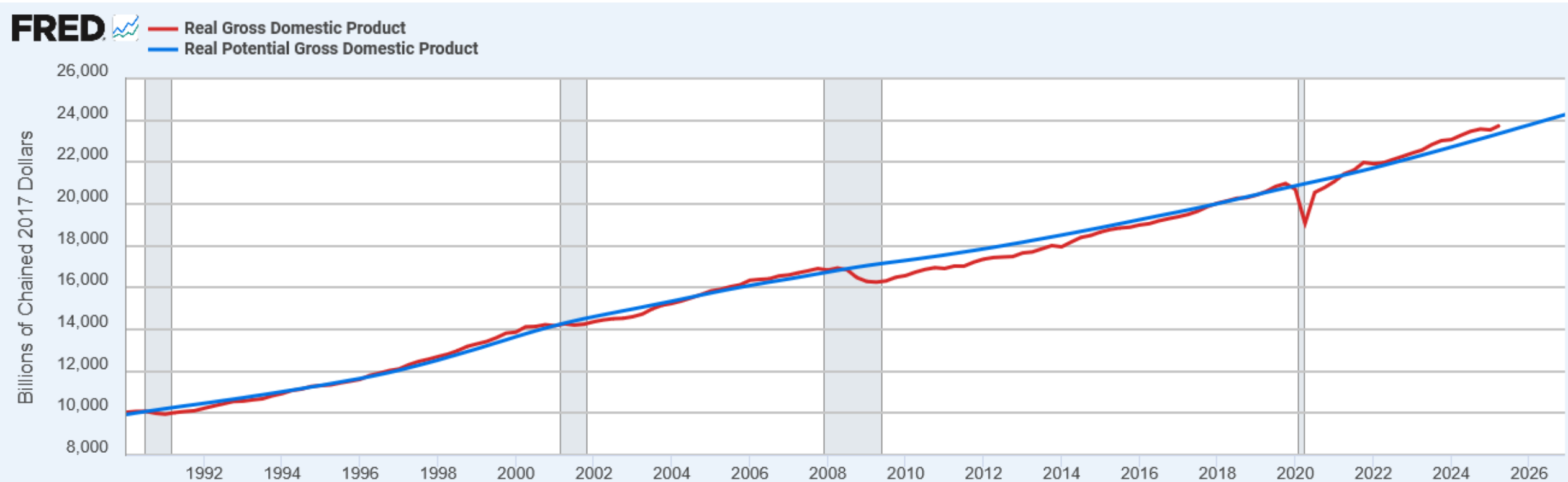


The Long-Run Aggregate-Supply Curve (*LRAS*)

- The **natural rate of output** (Y_N) is the amount of output the economy produces when unemployment is at its natural rate.
- Y_N is also called **potential output** or **full-employment output**.



Actual vs Potential Output

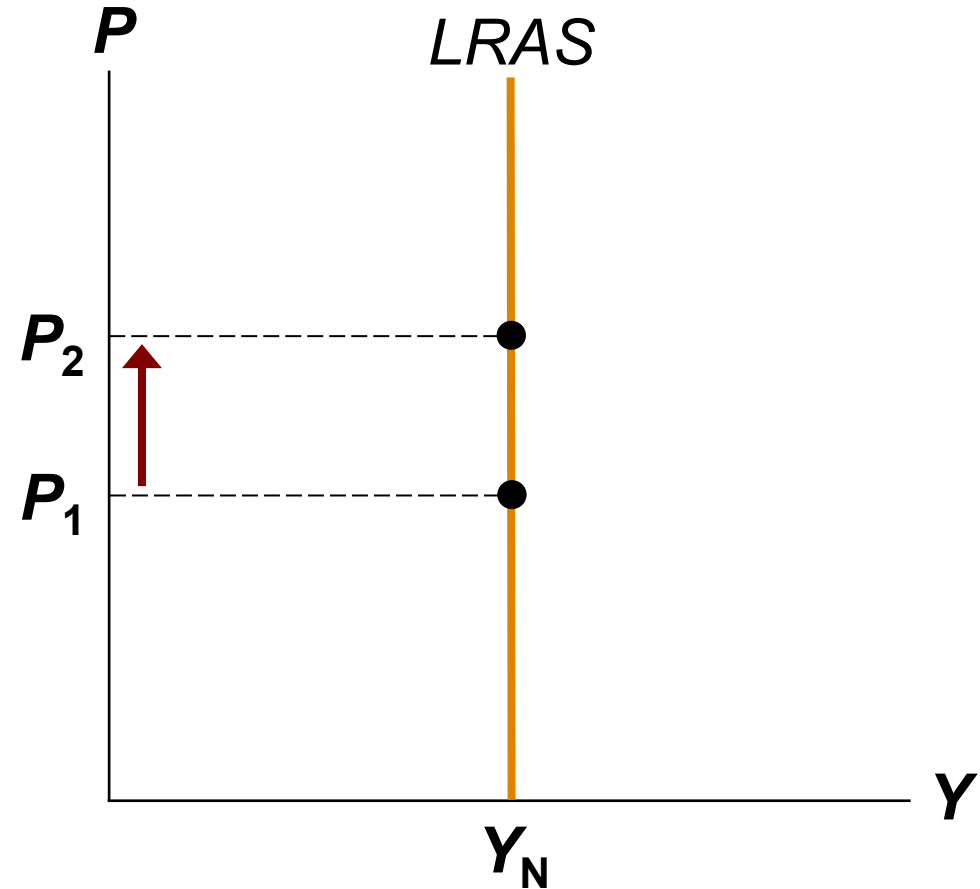


Sources: U.S. Bureau of Economic Analysis; U.S. Congressional Budget Office via FRED®
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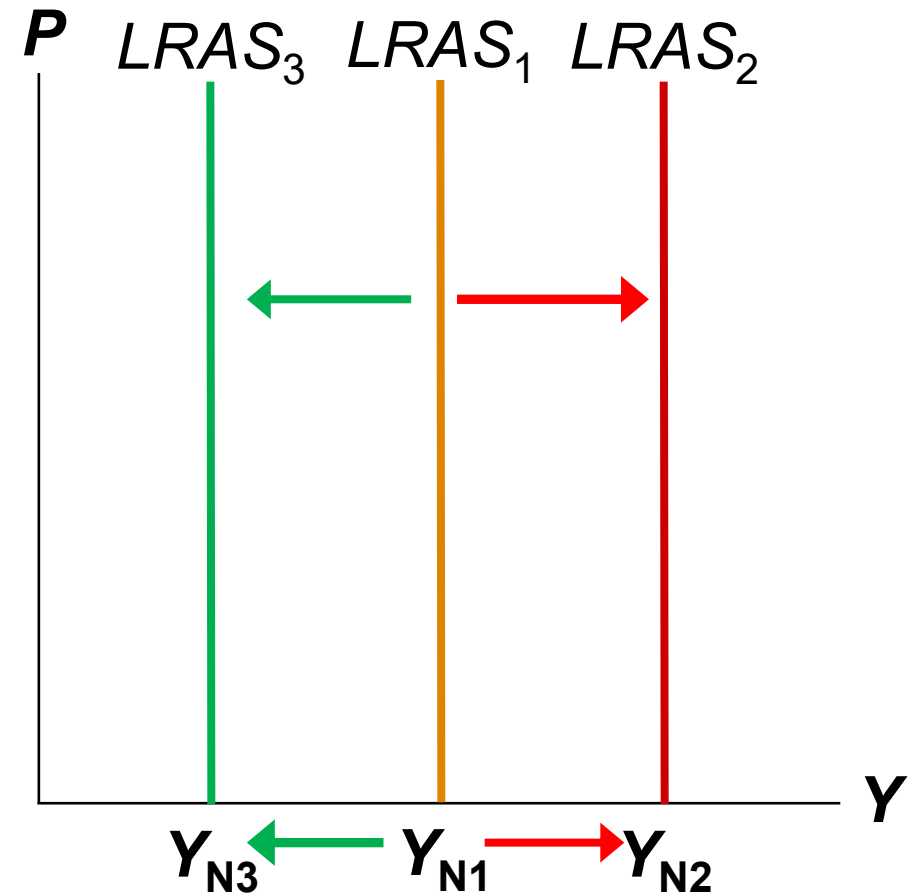
Why $LRAS$ Is Vertical

- Y_N determined by the economy's stocks of labor, capital, and natural resources, and on the level of technology.
- An increase in P does not affect any of these so, it does not affect Y_N .
 - This is the Classical Dichotomy



Why the $LRAS$ Curve Might Shift

- Any event that changes any of the determinants of Y_N will shift $LRAS$.
- Example: Immigration
 - increases L ,
 - causing Y_N to rise.
- Corruption in gov't causes changes in business environment causing Y_N to fall.



Why the *LRAS* Curve Might Shift

- Changes in *L* or natural rate of unemployment
 - Immigration (**Increase**)
 - Baby-boomers retire (**Decrease**)
 - Government policies reduce natural u-rate (**Increase**)
- Changes in *K* or *H*
 - Investment in factories, equipment (**Increase**)
 - More people get college degrees (**Increase**)
 - Factories destroyed by a hurricane (**Decrease**)

Why the *LRAS* Curve Might Shift

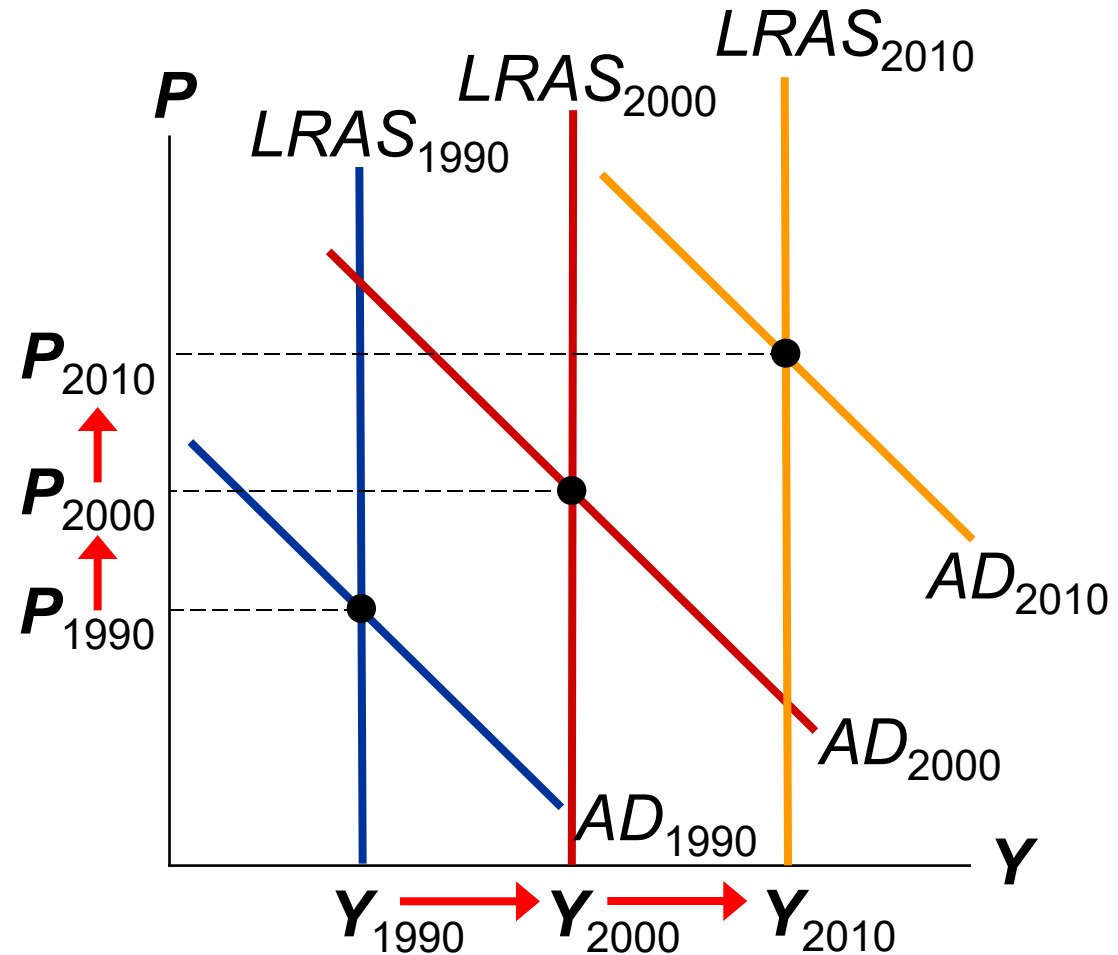
- Changes in natural resources
 - Discovery of new mineral deposits (**Increase**)
 - Reduction in supply of imported oil (**Decrease**)
 - Changing weather patterns that affect agricultural production (**Increase?**)
- Changes in technology
 - Productivity improvements from technological progress (**Increase**)

Using AD & AS to Depict Long-Run Growth and Inflation

Over the long run,
tech. progress shifts
 $LRAS$ to the right

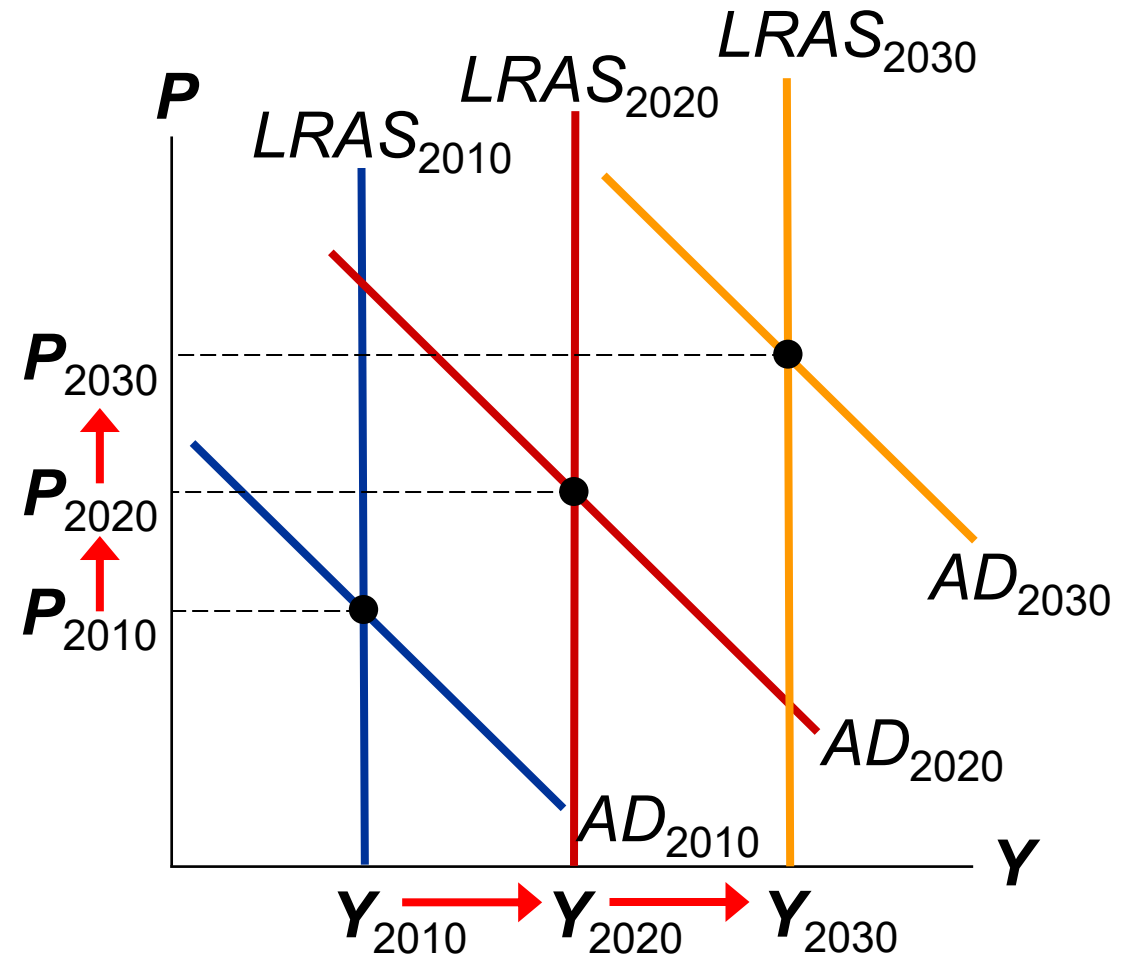
and growth in the
money supply shifts
 AD to the right.

Result:
ongoing inflation
and growth in
output.



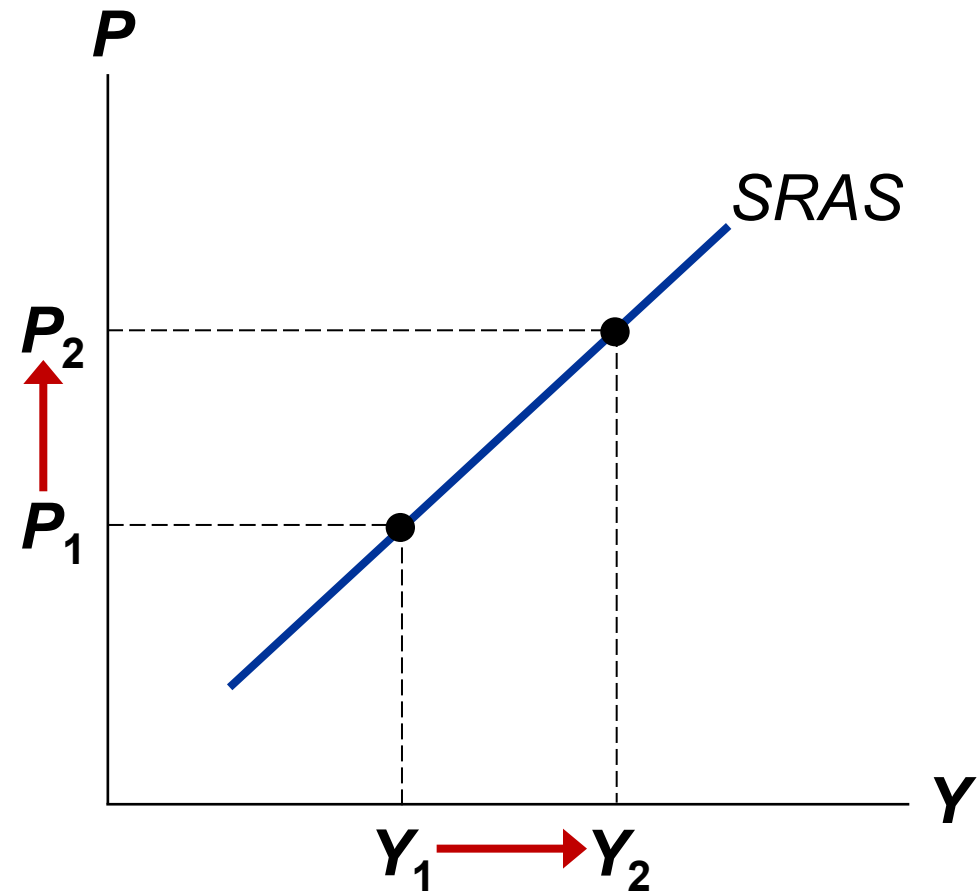
Using AD & AS to Depict Long-Run Growth and Inflation

- Over the long run, technological progress shifts $LRAS$ to the right
- And growth in the money supply shifts AD to the right.
- Result: ongoing inflation and growth in output.



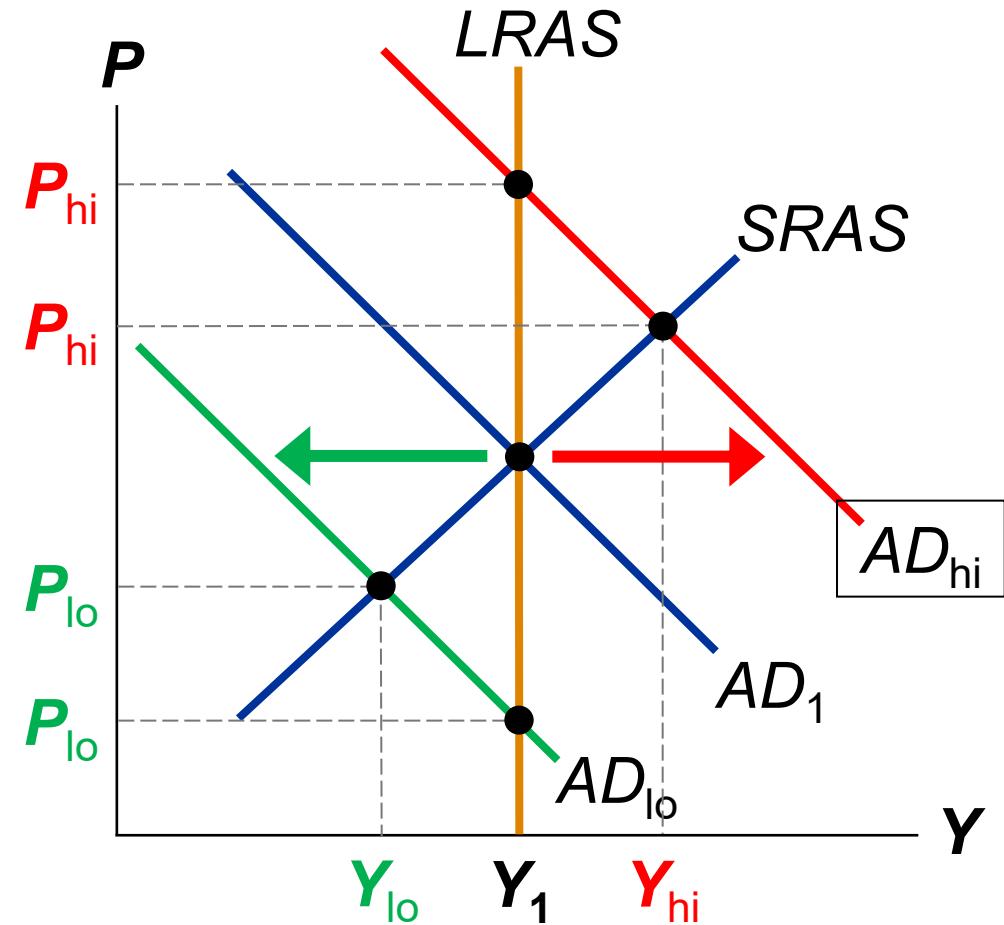
Short Run Aggregate Supply (*SRAS*)

- The *SRAS* curve is upward sloping:
- Over the period of 1–2 years, an increase in P causes an increase in the quantity of goods and services supplied.



Why the Slope of *SRAS* Matters

- If *AS* is vertical, fluctuations in *AD* do not cause fluctuations in output or employment.
- If *AS* slopes up, then shifts in *AD* do affect output and employment.



Three Theories of *SRAS*

In each,

- some type of market imperfection
- result:

Output deviates from its natural rate when the actual price level deviates from the price level people expected.

1. The Sticky-Wage Theory

- **Imperfection:**
Nominal wages are **sticky** in the short run, they adjust sluggishly.
 - Due to labor contracts, social norms
- Firms and workers set the nominal wage in advance based on P_E , the price level they expect to prevail.
- If $P > P_E$,
 - revenue is higher, but labor cost is not.
 - Production is more profitable, so firms increase output and employment.
- Hence, higher P causes higher Y , so the ***SRAS curve slopes upward***.

2. The Sticky-Price Theory

- **Imperfection:**

Many prices are sticky in the short run.

- Due to **menu costs**, the costs of adjusting prices.
- Examples: cost of printing new menus, the time required to change price tags

- Firms set sticky prices in advance based on P_E .

2. The Sticky-Price Theory cont.

- Suppose the Fed increases the money supply unexpectedly. In the long run, P will rise.
- In the short run, firms without menu costs can raise their prices immediately.
- Firms with menu costs wait to raise prices. Meanwhile, their prices are relatively low, which increases demand for their products, so they increase output and employment.
- Hence, higher P is associated with higher Y , so the *SRAS curve slopes upward*.

3. The Misperceptions Theory

- **Imperfection:**

Firms may confuse changes in P with changes in the relative price of the products they sell.

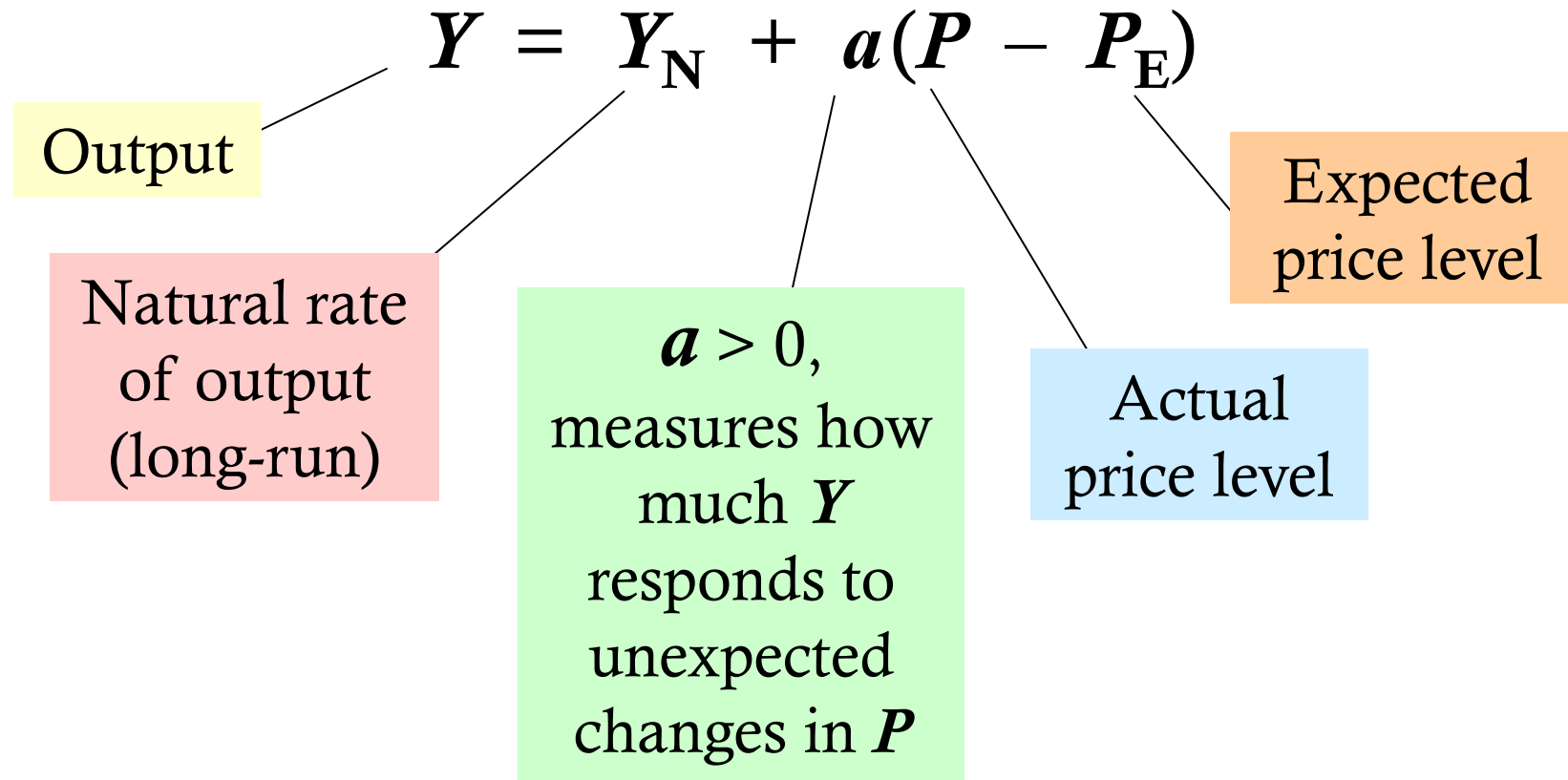
- If P rises above P_E , a firm sees its price rise before realizing all prices are rising.

- The firm may believe its *relative* price is rising and may increase output and employment.

- So, an increase in P can cause an increase in Y , making the ***SRAS curve upward-sloping.***

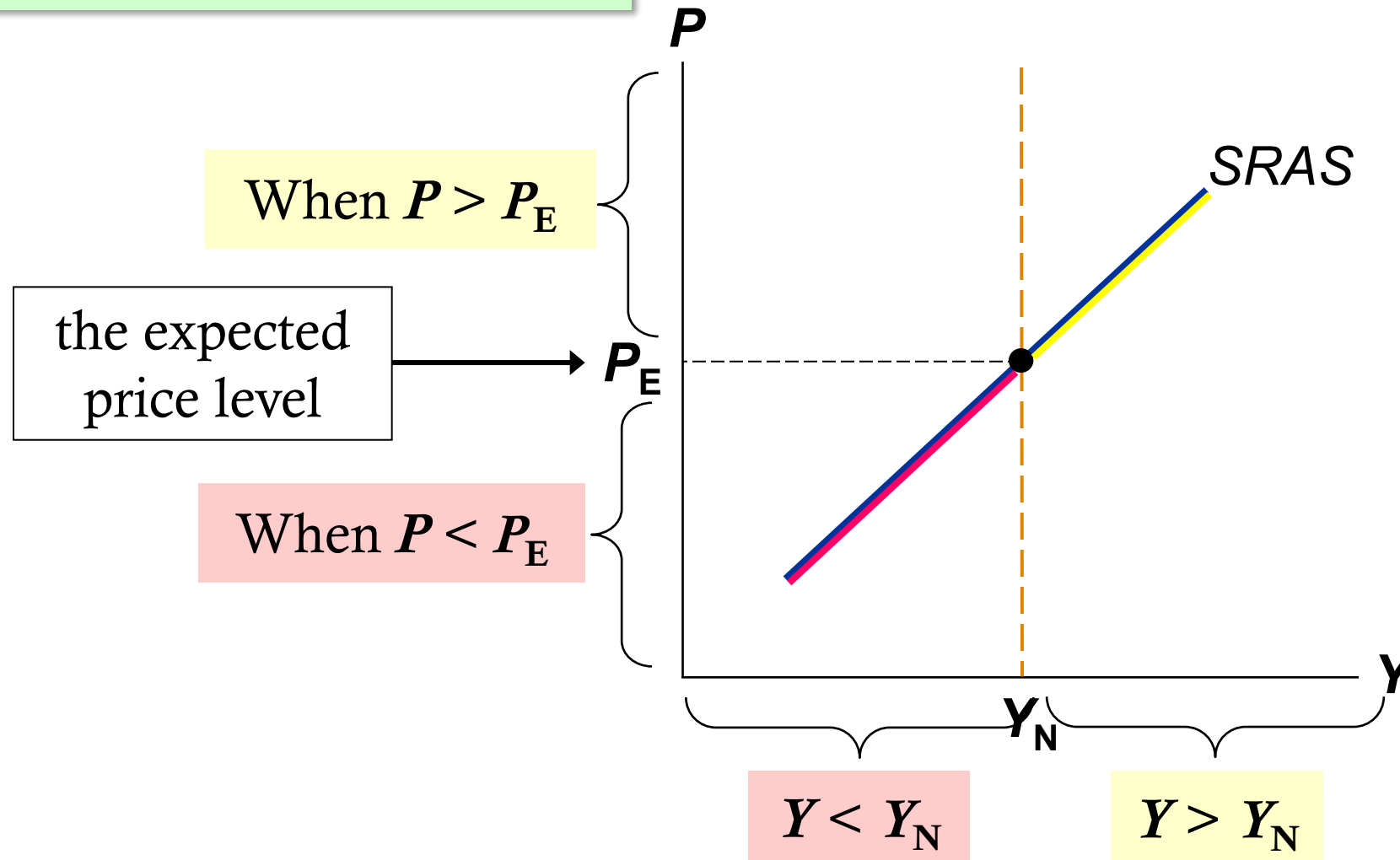
What the 3 Theories Have in Common:

- In all 3 theories, Y deviates from Y_N when P deviates from P_E .



What the 3 Theories Have in Common:

$$Y = Y_N + a(P - P_E)$$



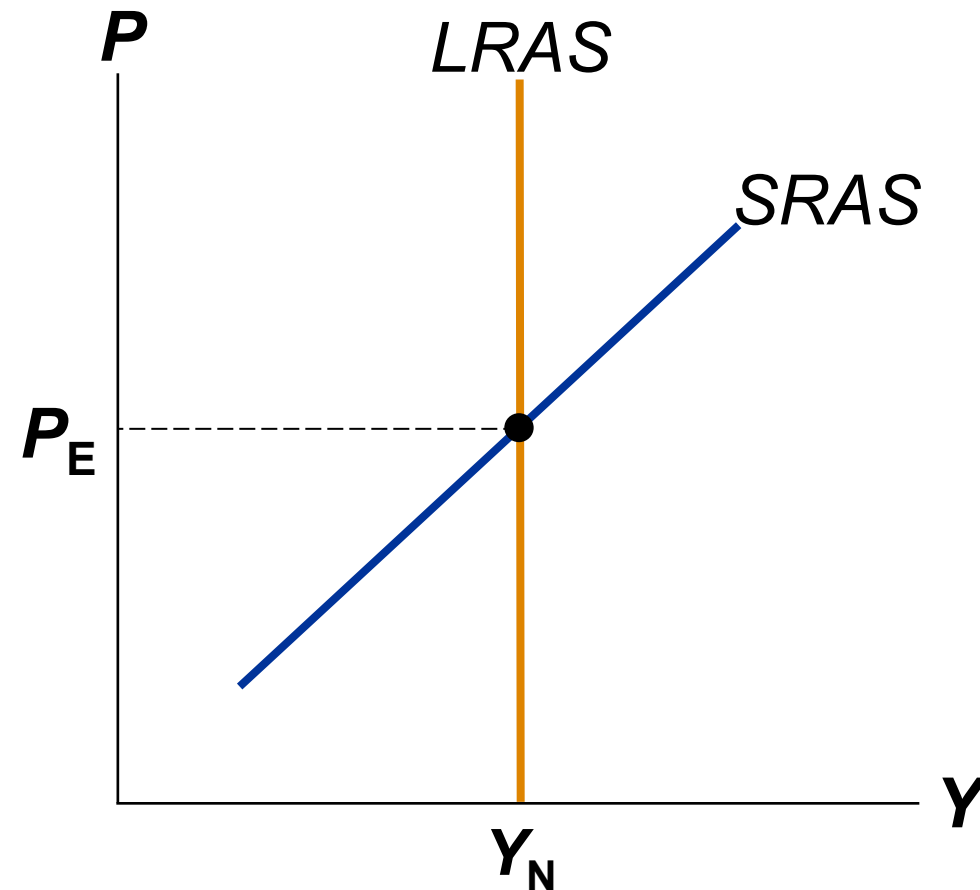
SRAS and *LRAS*

- The imperfections in these theories are temporary. Over time,
 - sticky wages and prices become flexible
 - misperceptions are corrected
- In the LR,
 - $P_E = P$
 - *AS* curve is vertical

SRAS and *LRAS*

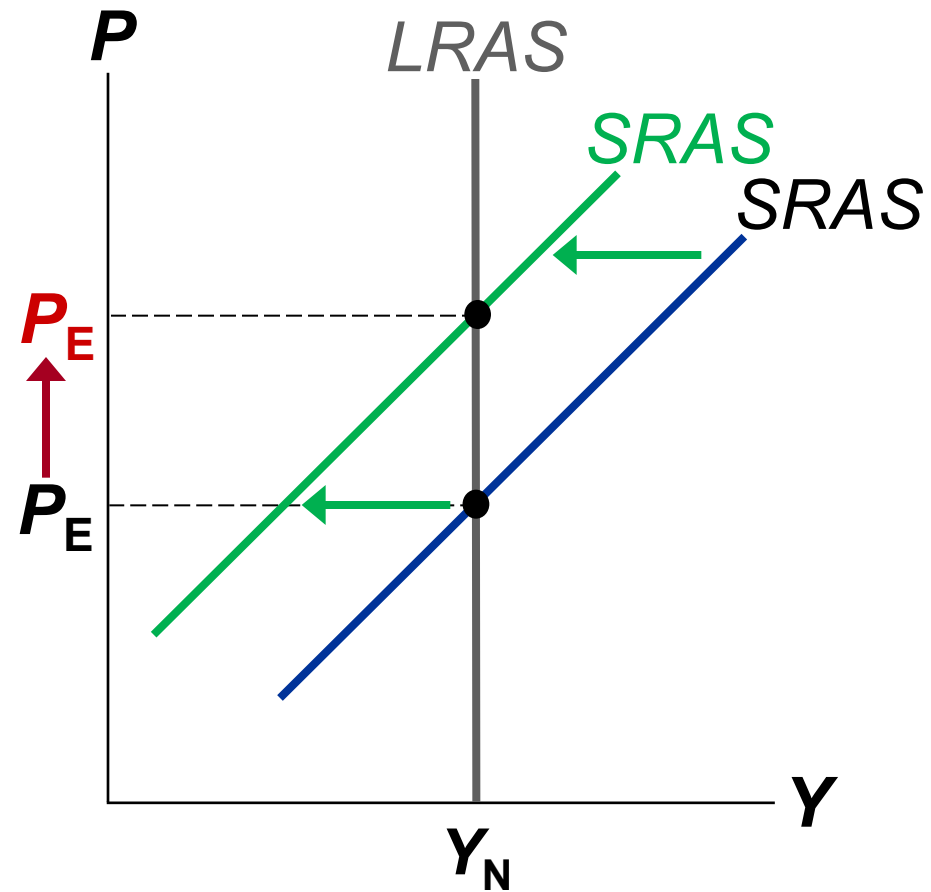
In the long run, $P_E = P$ and $Y = Y_N$.

$$Y = Y_N + a(P - P_E)$$



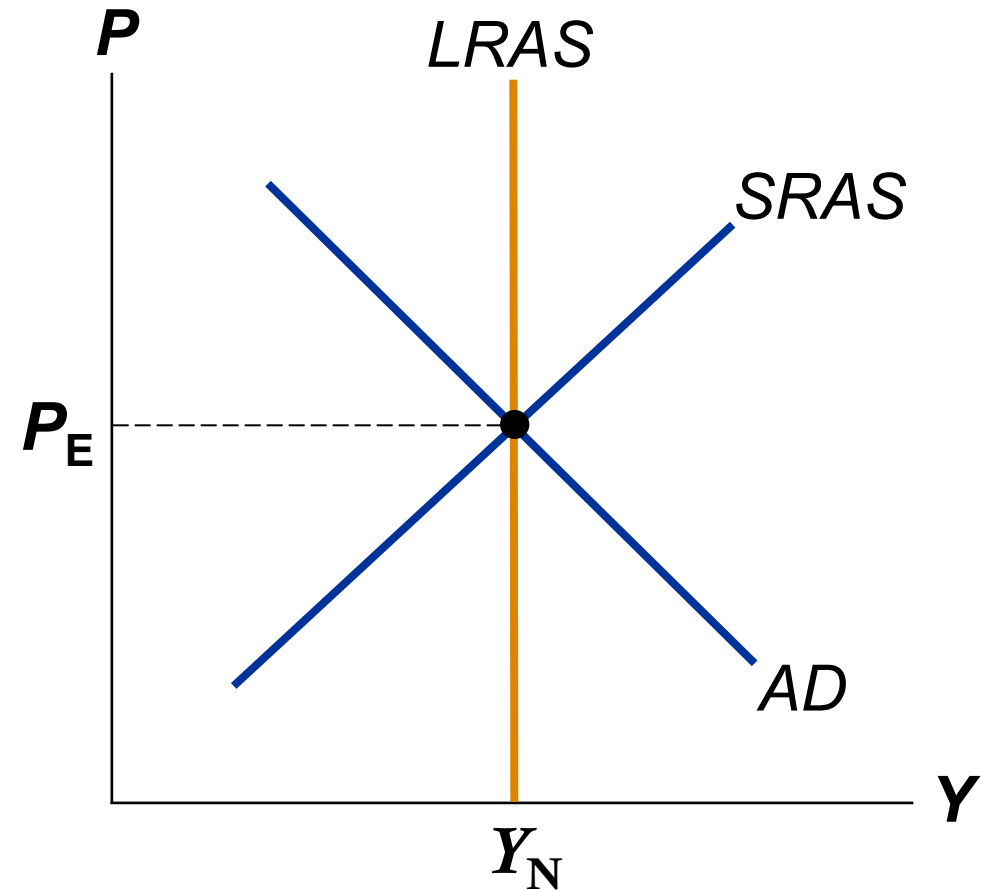
Why the *SRAS* Curve Might Shift

- Everything that shifts *LRAS* shifts *SRAS*, too.
- Also, P_E shifts *SRAS*:
- If P_E rises, workers & firms set higher wages.
- At each P , production is less profitable, Y falls, *SRAS* shifts left.



The Long-Run Equilibrium

- In the long-run equilibrium,
 - $P_E = P$,
 - $Y = Y_N$,
- and unemployment is at its natural rate.



Activity: What Shifts LRAS and SRAS?

- For each event below, decide whether it affects long-run aggregate supply (LRAS), short-run aggregate supply (SRAS), or both. Explain whether the curve shifts left or right, and why.

Event	Curve(s) Affected	Shift Direction	Why?
1. A wave of new immigrants enters the labor force.			
2. A severe hurricane destroys factories along the coast.			
3. The expected price level rises.			
4. A new technology improves productivity across industries.			

Activity: What Shifts LRAS and SRAS?

Event	Curve(s) Affected	Shift Direction	Explanation
1. A wave of new immigrants enters the labor force.	LRAS and SRAS	Right	More available labor raises potential output and short-run production capacity.
2. A severe hurricane destroys factories along the coast.	LRAS and SRAS	Left	Capital stock decreases, reducing both short-run and long-run productive capacity.
3. The expected price level rises.	SRAS only	Left	Higher expected prices lead firms and workers to demand higher nominal wages, reducing profitability at any given price level.
4. A new technology improves productivity across industries.	LRAS and SRAS	Right	Technological progress increases efficiency and output in both the short and long run.

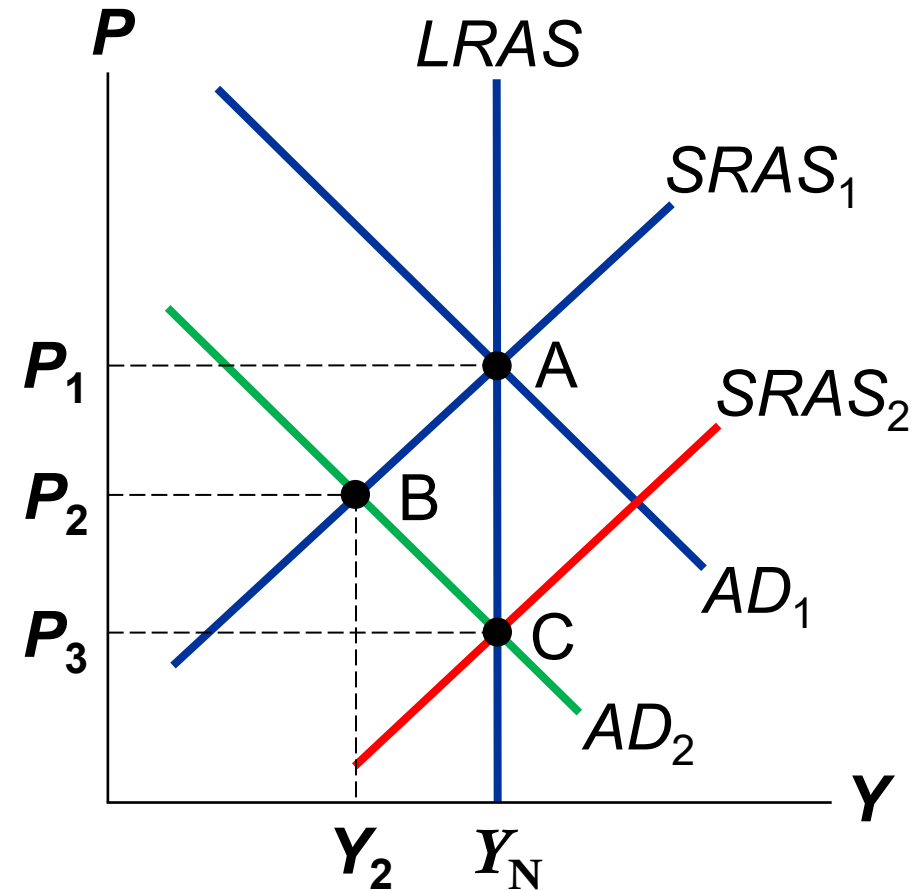
Activity: Effects of an Economic Event

- There is a crash in the stock market.
 1. Determine whether the event shifts AD or AS .
 2. Determine whether curve shifts left or right.
 3. Use $AD-AS$ diagram to see how the shift changes Y and P in the short run.
 4. Use $AD-AS$ diagram to see how economy moves from new SR equilibrium to new LR equilibrium.

Activity: Effects of an Economic Event

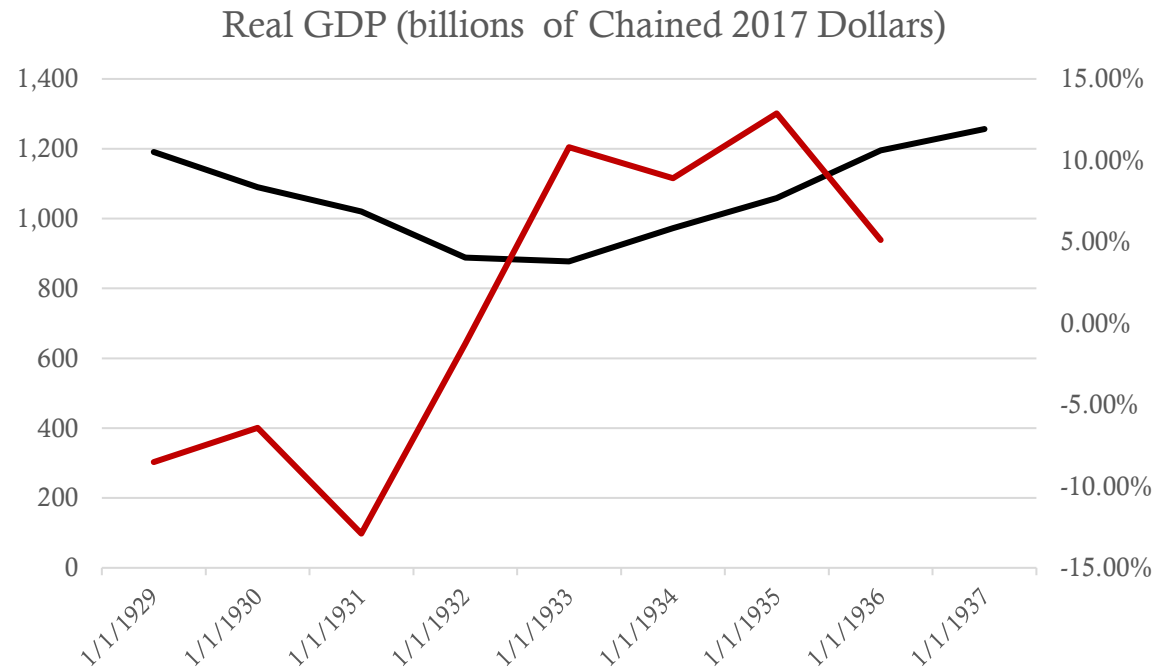
Event: Stock market crash

1. Affects C , AD curve
2. C falls, so AD shifts left
3. SR equilibrium at B.
 P and Y lower,
unemployment higher
4. Over time, P_E falls, $SRAS$
shifts right, until LR
equilibrium at C.
5. Y and unemployment back
at initial levels.



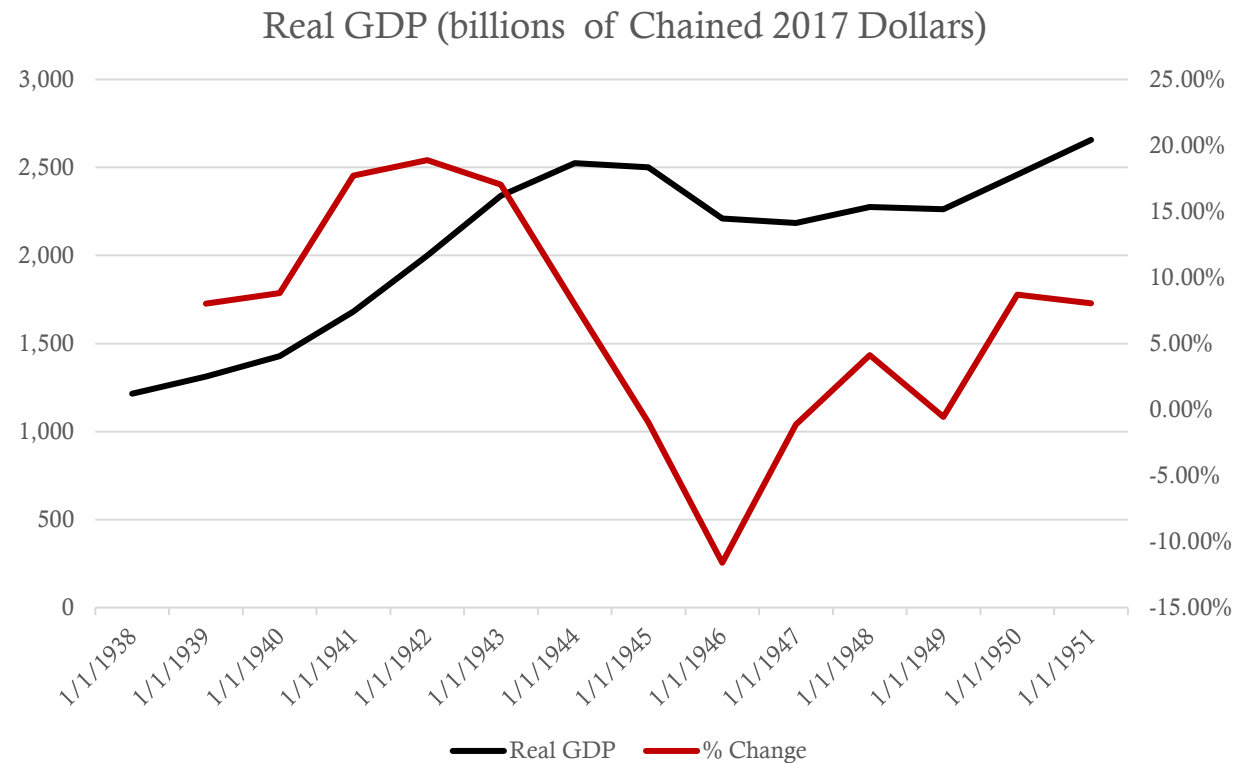
Two Big Shifts in Aggregate Demand

- **The Great Depression**
- From 1929–1933,
 - money supply fell 28% due to problems in banking system
 - stock prices fell 90%, reducing C and I
 - Y fell 27%
 - P fell 22%
 - u-rate rose from 3% to 25%



Two Big *AD* Shifts:

- **The World War II Boom**
- From 1939–1944,
 - govt outlays rose from \$9.1 billion to \$91.3 billion
 - Y rose 90%
 - P rose 20%
 - unemployment fell from 17% to 1%



Activity: Find the New Equilibrium

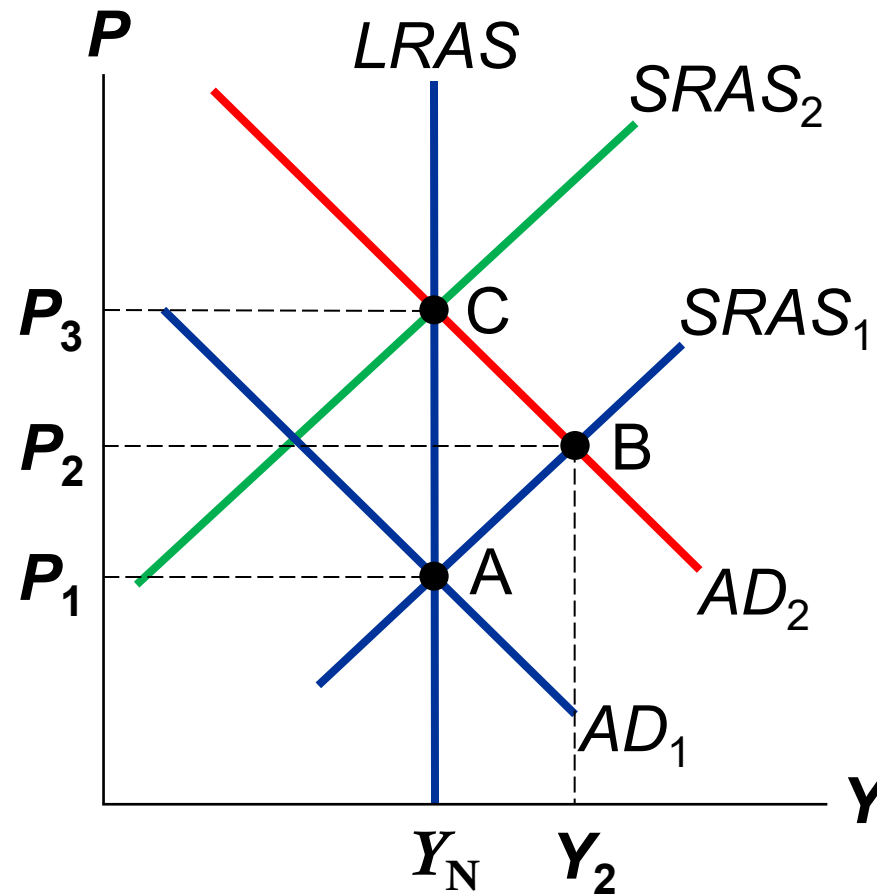
Draw the *AD-SRAS-LRAS* diagram for the U.S. economy starting in a long-run equilibrium.

1. A boom occurs in Canada. Use the AD/AS diagram to determine the SR and LR effects on U.S. GDP, the price level, and unemployment.
2. Oil prices surge causing rising costs for consumers and producers. Use the AD/AS diagram to determine the SR and LR effects on U.S. GDP, the price level, and unemployment.

Activity: Find the New Equilibrium

Event: Boom in Canada

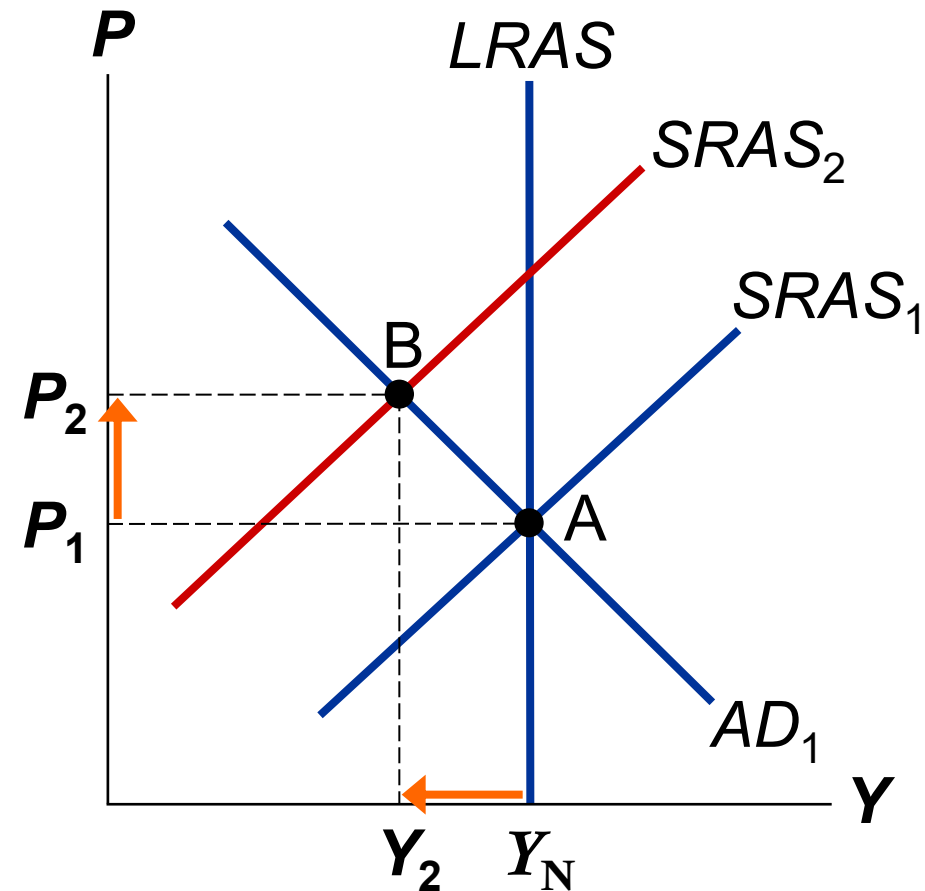
1. Affects NX , AD curve
2. Shifts AD right
3. SR equilibrium at point B.
 P and Y higher,
unemployment lower
4. Over time, P_E rises,
 $SRAS$ shifts left,
until LR equilibrium at C.
 Y and unemployment back
at initial levels.



Activity: Find the New Equilibrium

Event: Oil prices rise

1. Increases costs, shifts *SRAS* (assume *LRAS* constant)
2. *SRAS* shifts left
3. SR equilibrium at point B. P higher, Y lower, unemployment higher
4. From A to B, **stagflation**, a period of falling output and rising prices.



The 1970s Oil Shocks and Their Effects

	1973 – 1975	1978 – 1980
Real Oil Prices	+138%	+99%
CPI	+21%	+26%
Real GDP	-0.7%	+2.9%
Unemployed Persons	+3.5 million	+1.4 million

Key Takeaways

- The AD–AS model explains short-run fluctuations in output and prices.
- Aggregate demand captures total spending; aggregate supply captures total production.
- Short-run output can differ from potential GDP due to sticky prices and wages.
- Over time, the economy tends to return to its natural level of output.
- Policy actions that affect spending or production can stabilize or destabilize the economy.

Knowledge Check

1. What are the three effects that explain the downward slope of the aggregate demand curve?
2. What happens in the short run if aggregate demand shifts left?
3. Why is the long-run aggregate supply curve vertical?
4. How does the sticky-wage theory explain the upward slope of the short-run aggregate supply curve?
5. What combination of effects defines stagflation?

Knowledge Check Answers

1. The wealth effect, interest-rate effect, and exchange-rate effect.
2. Output and prices fall; unemployment rises.
3. Because output depends on real factors—labor, capital, resources, and technology—not the price level.
4. Wages adjust slowly, so unexpected changes in prices affect profitability and production.
5. Falling output and rising prices caused by a negative supply shock